Name ________________SOLUTIONS__________________

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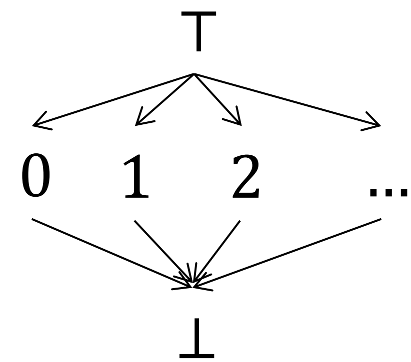
6.818 Fall 2020 | Miniquiz #26 | 5 Minutes

Assume you want to statically analyze the following MITScript VM bytecode program in order to determine the size of the operand stack at each point in the program:

```
1: if 3
2: load_const 0
3: load_const 1
4: call 3
5: neg
```



Furthermore, assume that stack sizes are represented using the flat lattice of integers as shown on the right.

1. For each type of instruction used in the program above, define a transfer function that computes the stack size right after the instruction is executed as a function of the stack size right before the instruction is executed (s_i):

   $T_{if\ m}(s_i) = s_i - 1$

   $T_{load\_const\ m}(s_i) = (s_i + 1)$ if $s_i \in \mathbb{Z}$ else s_i

   $T_{call\ m}(s_i) = (s_i - m + 1)$ if $s_i \in \mathbb{Z}$ else s_i

   $T_{neg}(s_i) = s_i$

2. Define a system of equations that represent the stack sizes s_i after each instruction i in the program above is executed, and then solve the system of equations. Assume that the stack initially contains two elements (i.e., $s_0 = 2$).

   $s_1 = T_{if\ 3}(s_0) = s_0 - 1 = 1$

   $s_2 = T_{load\_const\ 0}(s_1) = s_1 + 1 = 2$

   $s_3 = T_{load\_const\ 1}(s_2) = s_2 + 1 = 3$

   $s_4 = T_{call\ 3}(s_1 \sqcup s_3) = T_{call\ 3}(\top) = \top$

   $s_5 = T_{neg}(s_4) = \top$